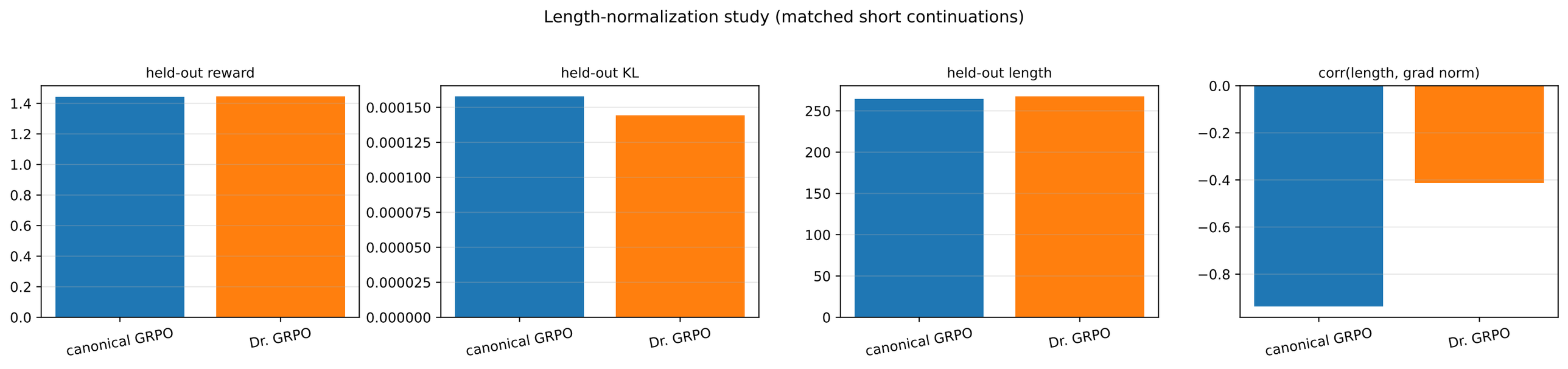
Length-normalization study (matched short continuations)
held-out reward
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
canonical GRPO
Dr. GRPO
held-out KL
0.000150
0.000125
0.000100
0.000075
0.000050
0.000025
0.000000
canonical GRPO
Dr. GRPO
held-out length
250
200
150
100
50
0
canonical GRPO
Dr. GRPO
corr(length, grad norm)
0.0
−0.2
−0.4
−0.6
−0.8
canonical GRPO
Dr. GRPO